👩‍🏫 Staff Guide · Biology A · Famous Fossils & Deep Time (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Ms. Dunham's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
A4.1 Washington's famous fossils (stations)	Reading parts 1 and 3; notes sections A and C
Rock layers game (AMNH); UCMP webquests	Reading part 2; notes section B
A4.2 Twine Time + Events in Evolution	Reading part 4; Twine Time scale math
A4.3 Building Bridges	Reading part 5; notes section C
Dinosaurs notebook check	Quick check + unit game

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Ms. Dunham's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Ms. Dunham.

🦕 Famous Fossils & Deep Time: Guided Notes

Biology A · Unit 7 · Level 3 (closer to the class work) · goes with Unit 4 Famous Fossils: A4.1 WA fossil stations, A4.2 Twine Time, A4.3 Building Bridges, rock layers game

Name: _____

Use the class slides and your reading. Key words:

fossil | *sediment* | *rock layers (strata)* | *extinct* | *imprint* | *petrified wood* | *deep time* | *land bridge* | *paleontologist*

A. How fossils form

1. The organism must be buried quickly in fine _____.
2. Over time, _____ replace bone, shell and wood.
3. Why are jellyfish fossils rare?

_____

_____

B. Rock layers and deep time (rock layers game, A4.2)

4. Deeper rock layers are _____.
5. Earth is about _____ billion years old.
6. **Twine Time scale**

> Example: if 4.6 billion years = 46 meters, then 1 meter = 100 million years

a) Life began 3.5 billion years ago. How many meters from today? _____

b) Dinosaurs died 66 million years ago. About how far from the end? _____

7. A species' column ends in layer C. What does that tell you?

_____

C. Washington fossils and land bridges (A4.1, A4.3)

8. Fill in the table.

Fossil	Where	Kind
Columbian mammoth	WA state fossil	
Blue Lake Rhino		mold in lava
Ginkgo forest	Vantage	

9. How did armadillos get to North America?

_____

_____

🦕 Famous Fossils & Deep Time: Reading

Biology A · Unit 7 · Level 3 (closer to the class work) · goes with Unit 4 Famous Fossils: A4.1 WA fossil stations, A4.2 Twine Time, A4.3 Building Bridges, rock layers game

Name: _____

Reading the Story in the Rocks

A fossil is the remains or trace of a living thing from long ago, kept in rock. For a fossil to form, the organism must be buried quickly in fine sediment like mud, sand or volcanic ash. Over a long time, minerals replace hard parts like bone, shell and wood. Some fossils are molds, imprints or footprints instead of the body itself.

Rock forms in layers called strata. Older layers are deeper and younger layers are on top. When scientists chart where a species appears, it shows up as a column through the layers. When the species goes extinct, its column ends.

Washington has famous fossils. The Columbian mammoth is the state fossil, chosen with help from elementary students. The first Washington dinosaur bone, from a theropod, was found in the San Juan Islands in 2012. Clovis stone tools found in East Wenatchee are about 11,000 to 12,000 years old. The Blue Lake Rhino is a mold in lava rock, and the Ginkgo Petrified Forest near Vantage has wood turned to stone.

Earth is about 4.6 billion years old, and life began about 3.5 billion years ago. In the Twine Time activity, a long piece of twine stands for all of Earth's history. Humans appear only at the very end, a tiny piece of the twine. This huge span is called deep time.

For millions of years North and South America were separate. Plate movement built the Central American land bridge, and animals crossed it: armadillos and opossums moved north. Paleontologists, scientists who study fossils, use these clues to learn how life changed.

Check your reading

1. What has to happen for a fossil to form?

2. In rock layers, where are the oldest fossils?

3. What is Washington's state fossil?

🦕 Famous Fossils & Deep Time: Quick Check

Biology A · Unit 7 · Level 3 (closer to the class work) · goes with Unit 4 Famous Fossils: A4.1 WA fossil stations, A4.2 Twine Time, A4.3 Building Bridges, rock layers game

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Fossils form best when an organism is…

A. Buried quickly in fine sediment | B. Left on top of the ground | C. Frozen for one day

2. The oldest rock layers are…

A. At the top | B. In the middle | C. At the bottom

3. Earth is about…

A. 6,000 years old | B. 1 million years old | C. 4.6 billion years old

4. Petrified wood forms when…

A. Wood melts | B. Minerals replace wood | C. Wood is painted

5. Explain what the Twine Time activity shows about humans.

6. Name one famous Washington fossil and what it is.

🦕 Famous Fossils & Deep Time: Guided Notes · ANSWER KEY

Biology A · Unit 7 · Level 3 (closer to the class work) · goes with Unit 4 Famous Fossils: A4.1 WA fossil stations, A4.2 Twine Time, A4.3 Building Bridges, rock layers game

Name: _____

Use the class slides and your reading. Key words:

fossil · *sediment* · *rock layers (strata)* · *extinct* · *imprint* · *petrified wood* · *deep time* · *land bridge* · *paleontologist*

A. How fossils form

1. The organism must be buried quickly in fine **sediment**.
2. Over time, **minerals** replace bone, shell and wood.
3. Why are jellyfish fossils rare?
Jellyfish have no hard parts for minerals to replace.

B. Rock layers and deep time (rock layers game, A4.2)

4. Deeper rock layers are **older**.
5. Earth is about **4.6** billion years old.
6. **Twine Time scale**

> Example: if 4.6 billion years = 46 meters, then 1 meter = 100 million years

a) Life began 3.5 billion years ago. How many meters from today? **35 meters**

b) Dinosaurs died 66 million years ago. About how far from the end? **about 0.66 m (66 cm)**

7. A species' column ends in layer C. What does that tell you?
The species went extinct at the time layer C formed.

C. Washington fossils and land bridges (A4.1, A4.3)

8. Fill in the table.

Fossil	Where	Kind
Columbian mammoth	WA state fossil	**bones**
Blue Lake Rhino	**Blue Lake / Grand Coulee**	mold in lava
Ginkgo forest	Vantage	**petrified wood**

9. How did armadillos get to North America?
They walked north across the Central American land bridge formed by plate movement.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

🦕 Famous Fossils & Deep Time: Reading · ANSWER KEY

Biology A · Unit 7 · Level 3 (closer to the class work) · goes with Unit 4 Famous Fossils: A4.1 WA fossil stations, A4.2 Twine Time, A4.3 Building Bridges, rock layers game

Name: _____

Reading the Story in the Rocks

A fossil is the remains or trace of a living thing from long ago, kept in rock. For a fossil to form, the organism must be buried quickly in fine sediment like mud, sand or volcanic ash. Over a long time, minerals replace hard parts like bone, shell and wood. Some fossils are molds, imprints or footprints instead of the body itself.

Rock forms in layers called strata. Older layers are deeper and younger layers are on top. When scientists chart where a species appears, it shows up as a column through the layers. When the species goes extinct, its column ends.

Washington has famous fossils. The Columbian mammoth is the state fossil, chosen with help from elementary students. The first Washington dinosaur bone, from a theropod, was found in the San Juan Islands in 2012. Clovis stone tools found in East Wenatchee are about 11,000 to 12,000 years old. The Blue Lake Rhino is a mold in lava rock, and the Ginkgo Petrified Forest near Vantage has wood turned to stone.

Earth is about 4.6 billion years old, and life began about 3.5 billion years ago. In the Twine Time activity, a long piece of twine stands for all of Earth's history. Humans appear only at the very end, a tiny piece of the twine. This huge span is called deep time.

For millions of years North and South America were separate. Plate movement built the Central American land bridge, and animals crossed it: armadillos and opossums moved north. Paleontologists, scientists who study fossils, use these clues to learn how life changed.

Check your reading

1. What has to happen for a fossil to form?
The organism is buried quickly in fine sediment, and minerals slowly replace hard parts.
2. In rock layers, where are the oldest fossils?
In the deepest (bottom) layers.
3. What is Washington's state fossil?
The Columbian mammoth.

🦕 Famous Fossils & Deep Time: Quick Check · ANSWER KEY

Biology A · Unit 7 · Level 3 (closer to the class work) · goes with Unit 4 Famous Fossils: A4.1 WA fossil stations, A4.2 Twine Time, A4.3 Building Bridges, rock layers game

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Fossils form best when an organism is…

A. Buried quickly in fine sediment | B. Left on top of the ground | C. Frozen for one day

2. The oldest rock layers are…

A. At the top | B. In the middle | **C. At the bottom**

3. Earth is about…

A. 6,000 years old | B. 1 million years old | **C. 4.6 billion years old**

4. Petrified wood forms when…

A. Wood melts | **B. Minerals replace wood** | C. Wood is painted

5. Explain what the Twine Time activity shows about humans.

Humans appear only at the very end; we are a tiny part of Earth's long history.

6. Name one famous Washington fossil and what it is.

Any: Columbian mammoth (state fossil), Blue Lake Rhino (mold), theropod bone (first WA dinosaur), Ginkgo petrified wood, Clovis tools.

🔑 *Key written by our team from Ms. Dunham's quizzes and notes, please check.*